ESP32-S3-DevKitC-1

3V3 pin → sensor VCC.
GND → sensor GND.
GPIO4 ↔ sensor signal S.
No PLC 24V connection.

3V3 → VCC

GPIO ↔ S

GND → GND

SEN0137 / DHT22

Actual VCC 3.3V: user confirmed 2026-10-13.
S direct to GPIO4.
Room-condition readings passed user tests.

INTERFACE BASIS

Manufacturer lists VCC 5V / output 0..3.3V.
Current tested wiring is 3V3, not assumed 5V.
Confirm sensor operating supply range before final release.
Do not raise VCC blindly; verify signal/pull-up voltage.

FC04 INPUT REGISTERS / TCP502 / UNIT1			
Offset	Raw field	Type	Meaning
0	MAP_VERSION	U16	1
1	BOOT_ID	U16	Boot identity 1..32000
2	SAMPLE_SEQ	U16	Initial 0; new good sample 1..32000
3	TEMP_X100	I16	Degrees C ×100
4	RH_X100	U16	%RH ×100
5	VALID	U16	1 valid /0 invalid
6	STATUS	U16	0 OK /1 wait /2 read /3 CRC /4 range /5 stale
7	AGE_100MS	U16	Time since last good measurement /100ms

PACKED EXTENSION / OFFSET 8..23

Eight DWORDs, high-word first.
DWORDi = generation<<16 | raw_i.
Registers 8+2i=generation, 9+2i=raw_i.
Read all eight items with one same generation.
Generation 1..32000; wrap supported.

SAMPLING

GPIO4 driver interval 2500ms.
Raw sample stale after 6s.
Last numbers may remain when invalid: check VALID/STATUS.

Sensor source: actual native firmware + user wiring report; supply-range issue remains OPEN. Wi-Fi / IP configuration is not a PLC hardware port.

NO.	DATE	REVISION
R0	2026-10-22	FUNCTION RECORD

DWG. TITLE

SEMICON SORTER PROJECT
Final Test and Multi-Bin Sorting Cell

ESP32 SENSOR WIRING AND DATA FRAME

DRAWN BY	DATE
J. H. KO	2026-10-14

DWG. NO
010

SHEET NO
1 OF 1